

电催化计算实战入门

VASP实操1—吸附能计算

主讲人：高怡静

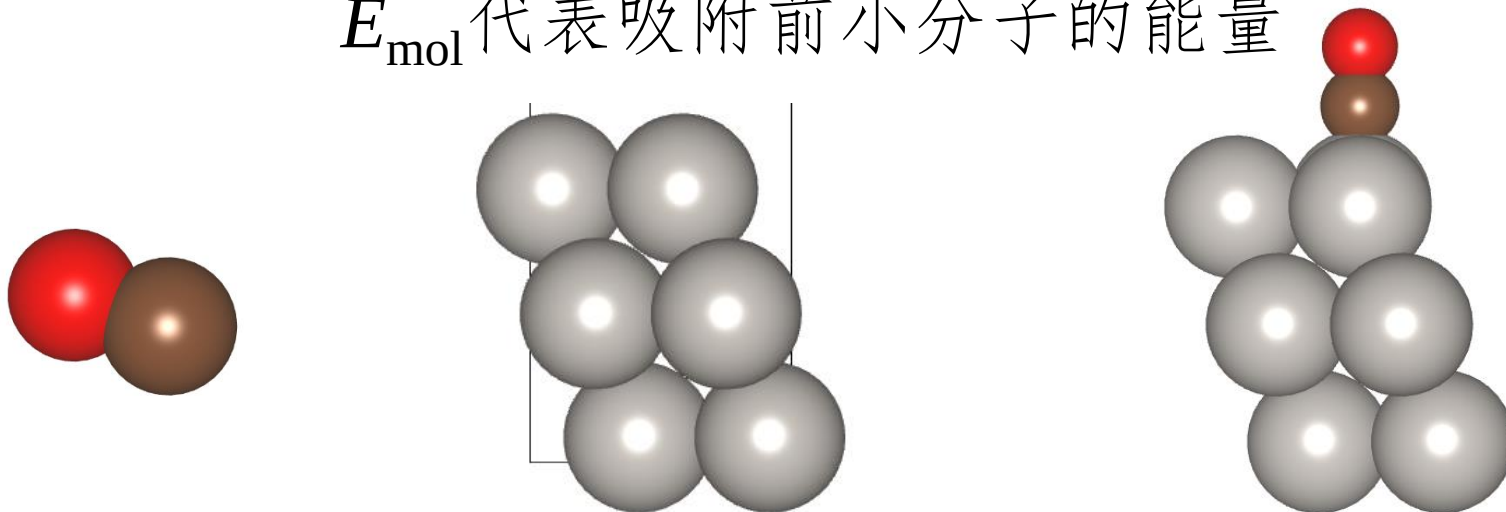
计算公式：

$$E_{\text{ad}} = E_{\text{total}} - (E_{\text{sub}} + E_{\text{mol}})$$

E_{total} 代表吸附后整个体系的能量

E_{sub} 代表吸附前催化剂的能量

E_{mol} 代表吸附前小分子的能量



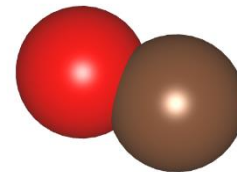
1.CO结构优化

```
SYSTEM =CO

#Startparameter for this run:
PREC      = A
ISTART    = 0
ICHARG    = 2
ENCUT     = 500
NELM      = 1000; #maximum of e
EDIFF     = 1E-5
LREAL     = Auto
EDIFFG    = -1E-2
NSW       = 1000  # number of io
IBRION    = 2    # optimization
ISIF      = 2
ISMEAR    =0
SIGMA     = 0.2
ISYM=0
LCHARG=F
LWAVE=F
GGA=PE
```

Automatic mesh

```
0
G
3   3   3
0.  0.  0.
```



2.Pt结构优化

```
SYSTEM =CO

#Startparameter for this run:
PREC    = A
ISTART   = 0
ICHARG   = 2
ENCUT    = 500
NELM     = 1000; #maximum of e.
EDIFF    = 1E-5
LREAL    = Auto
EDIFFG   = -1E-2
NSW      = 1000  # number of io:
IBRION    = 2    # optimization
ISIF      = 2
ISMEAR   =0
SIGMA    = 0.2
ISYM=0
LCHARG=F
LWAVE=F
GGA=PE
```

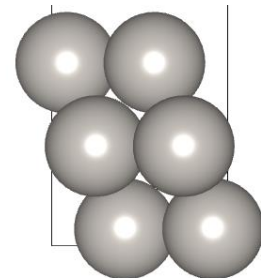
Automatic mesh

0

G

3 3 1

0. 0. 0.



3. CO-Pt 结构优化

```
SYSTEM =CO

#Startparameter for this run:
PREC      = A
ISTART    = 0
ICHARG    = 2
ENCUT     = 500
NELM      = 1000; #maximum of e.
EDIFF     = 1E-5
LREAL     = Auto
EDIFFG    = -1E-2
NSW       = 1000  # number of io:
IBRION    = 2    # optimization
ISIF      = 2
ISMEAR    =0
SIGMA     = 0.2
ISYM=0
LCHARG=F
LWAVE=F
GGA=PE
```

Automatic mesh

0

G

3

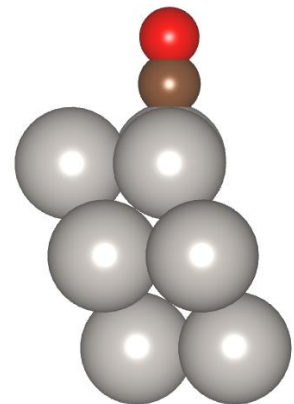
3

1

0.

0.

0.



4. 查看能量带入公式计算

CO能量

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free energy TOTEN = -14.77733834 eV
energy without entropy= -14.77733834 energy(sigma->0) = -14.77733834
```

Pt能量

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free energy TOTEN = -74.65180675 eV
energy without entropy= -74.12680264 energy(sigma->0) = -74.38930469
d Force = 0.1258602E-02[ 0.256E-04, 0.249E-02] d Energy = 0.1256818E-02 0.178E-05
d Force =-0.6563931E+02[-0.657E+02,-0.656E+02] d Ewald =-0.6563933E+02 0.180E-04
```

CO-Pt能量

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free energy TOTEN = -91.44310276 eV
energy without entropy= -91.05533254 energy(sigma->0) = -91.24921765
d Force = 0.5476150E-05[ 0.498E-05, 0.597E-05] d Energy = 0.6147340E-05-0.671E-06
d Force = 0.2403974E+00[ 0.240E+00, 0.240E+00] d Ewald = 0.2403974E+00-0.498E-09
```

该结构下，CO在Pt(111)表面的吸附能：

$$E_{\text{ad}} = -91.24921765 - (-14.77733834 + (-74.38930469)) = -2.08 \text{ eV}$$

电催化计算实战入门

VASP实操2—单点能计算

主讲人：高怡静

单点能计算

单点能计算：对给定几何构型的体系进行能量或者性质计算。往往是在结构优化的基础上进行。除了**INCAR**设置有区别外，将结构优化后的**CONTCAR**改成**POSCAR**；**KPOINTS**和**POTCAR**不变。

```
SYSTEM =Pt

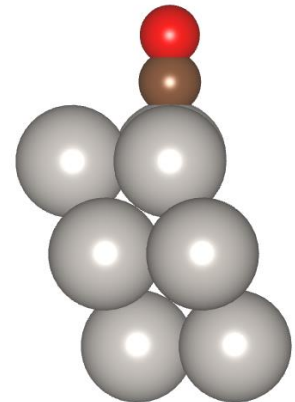
#Startparameter for this run:
PREC   = A
ISTART = 0
ICHARG = 2
ISPIN=2

ENCUT   = 500
NELM    = 1000; #maximum of electronic step
EDIFF   = 1E-5
LREAL   = Auto
EDIFFG  = -1E-2
#NSW    = 1000 # number of ionic step
IBRION  = -1 # optimization
ISIF    = 2
ISMEAR  = 0
SIGMA   = 0.2
ISYM=0
LCHARG=F
LWAVE=F
GGA=PE
#IALGO  = 48
IVDW=11
```

单点能计算：离子步的最大数目NSW设为0

Automatic mesh

```
0
G
3 3 1
0. 0. 0.
```



单点能计算

计算完成后在输出文件**OUTCAR**中查看单点能。

CO-Pt单点能计算结果

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free energy TOTEN = -91.44309494 eV
energy without entropy= -91.05528681 energy(sigma->0) = -91.24919088
```

CO-Pt结构优化计算结果

```
FREE ENERGIE OF THE ION-ELECTRON SYSTEM (eV)
-----
free energy TOTEN = -91.44310276 eV
energy without entropy= -91.05533254 energy(sigma->0) = -91.24921765
```

电催化计算实战入门

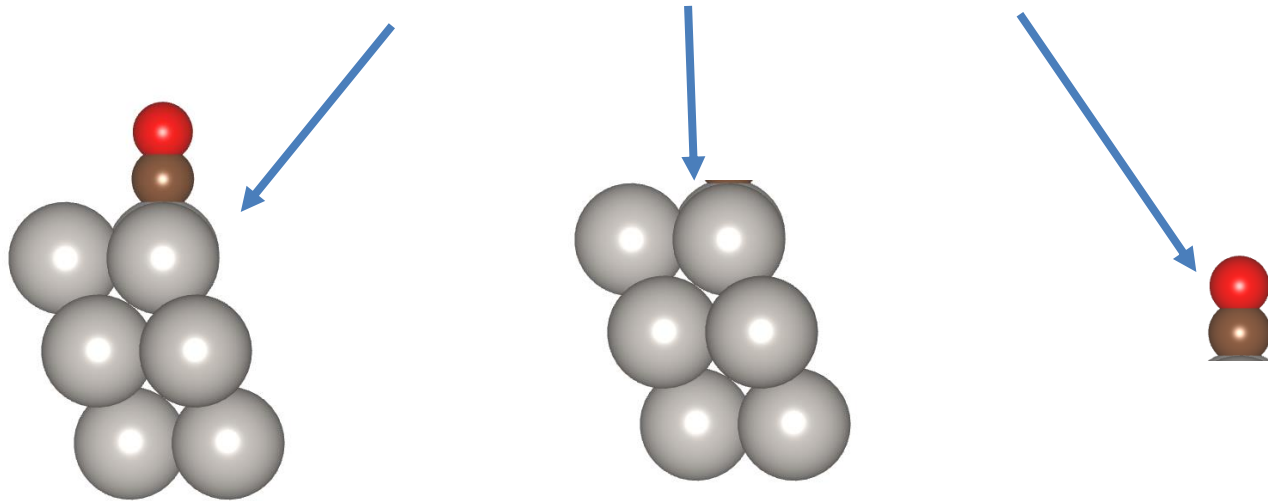
VASP实操3—差分电荷密度计算

主讲人：高怡静

差分电荷密度计算的前提是完成了结构优化计算。

差分电荷计算公式：

$$\Delta\rho = \rho_{AB} - \rho_A - \rho_B$$



注：A、B是优化后结构的一部分，与吸附能计算有区别

1. 计算CO-Pt的电荷密度

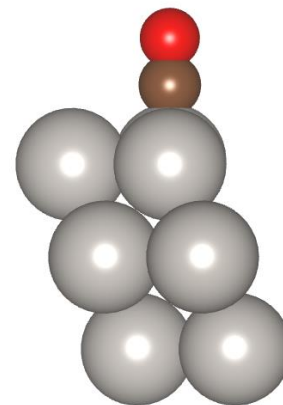
```
SYSTEM =Pt

#Startparameter for this run:
PREC    = A
ISTART  = 0
ICHARG  = 2

ENCUT   = 500
NELM    = 1000; #maximum of electroni
EDIFF   = 1E-5
LREAL   = Auto
EDIFFG  = -1E-2
NSW     = 0  # number of ioinc step
IBRION  = -1  # optimization
ISIF    = 2
ISMEAR  =0
SIGMA   = 0.2
LCHARG=T
LWAVE=T
GGA=PE
IVDW=11
```

Automatic mesh

```
0
G
3   3   1
0.  0.  0.
```



2. 计算Pt的电荷密度

```
SYSTEM =Pt

#Startparameter for this run:
PREC    = A
ISTART  = 0
ICHARG  = 2

ENCUT   = 500
NELM    = 1000; #maximum of electronic step
EDIFF   = 1E-5
LREAL   = Auto
EDIFFG  = -1E-2
NSW     = 0    # number of ioinc step
IBRION  = -1   # optimization
ISIF    = 2
ISMear  =0
SIGMA   = 0.2
LCHARG=T
LWAVE=T
GGA=PE
IVDW=11
```

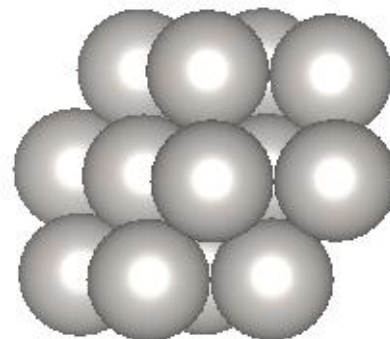
Automatic mesh

0

G

3 3 1

0. 0. 0.

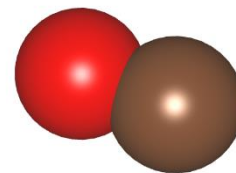


3. 计算CO的电荷密度

```
SYSTEM =Pt  
  
#Startparameter for this run:  
PREC   = A  
ISTART = 0  
ICHARG = 2  
  
ENCUT  = 500  
NELM   = 1000; #maximum of electronic step  
EDIFF  = 1E-5  
LREAL  = Auto  
EDIFFG = -1E-2  
NSW    = 0   # number of ioinc step  
IBRION  = -1  # optimization  
ISIF    = 2  
ISMEAR  =0  
SIGMA   = 0.2  
LCHARG=T  
LWAVE=T  
GGA=PE  
IVDW=11
```

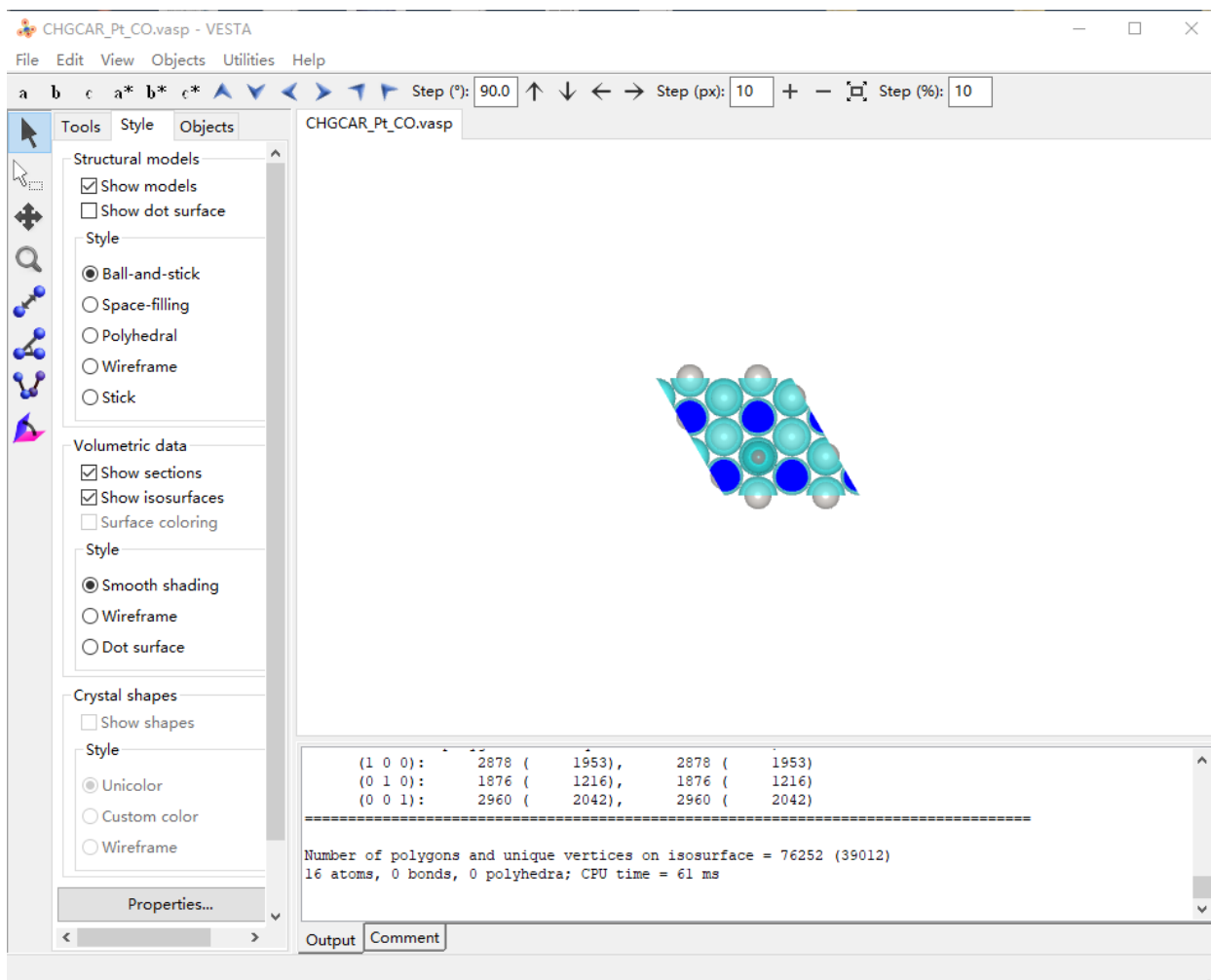
Automatic mesh

```
0  
G  
3   3   1  
0.  0.  0.
```

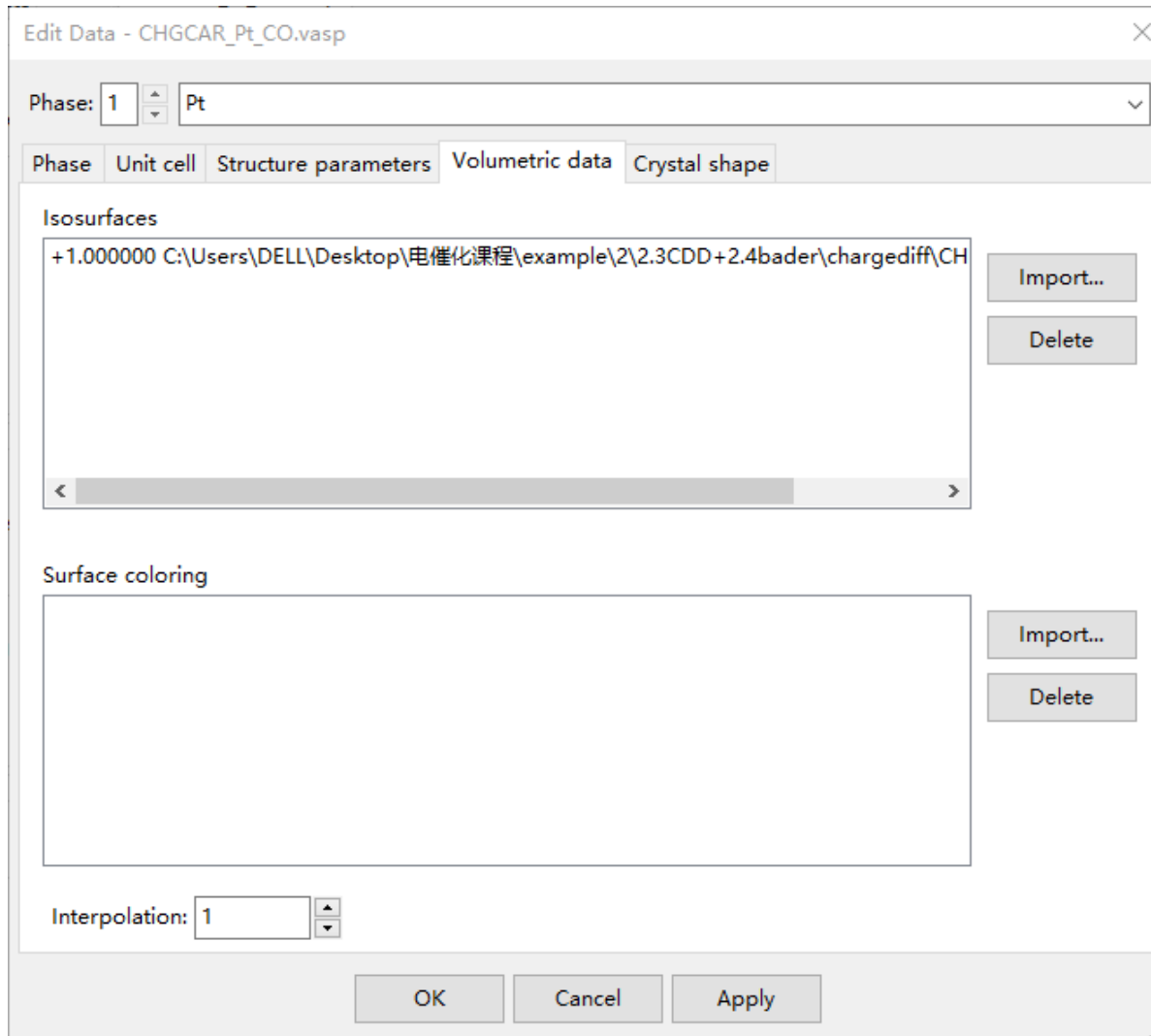


差分电荷密度可视化

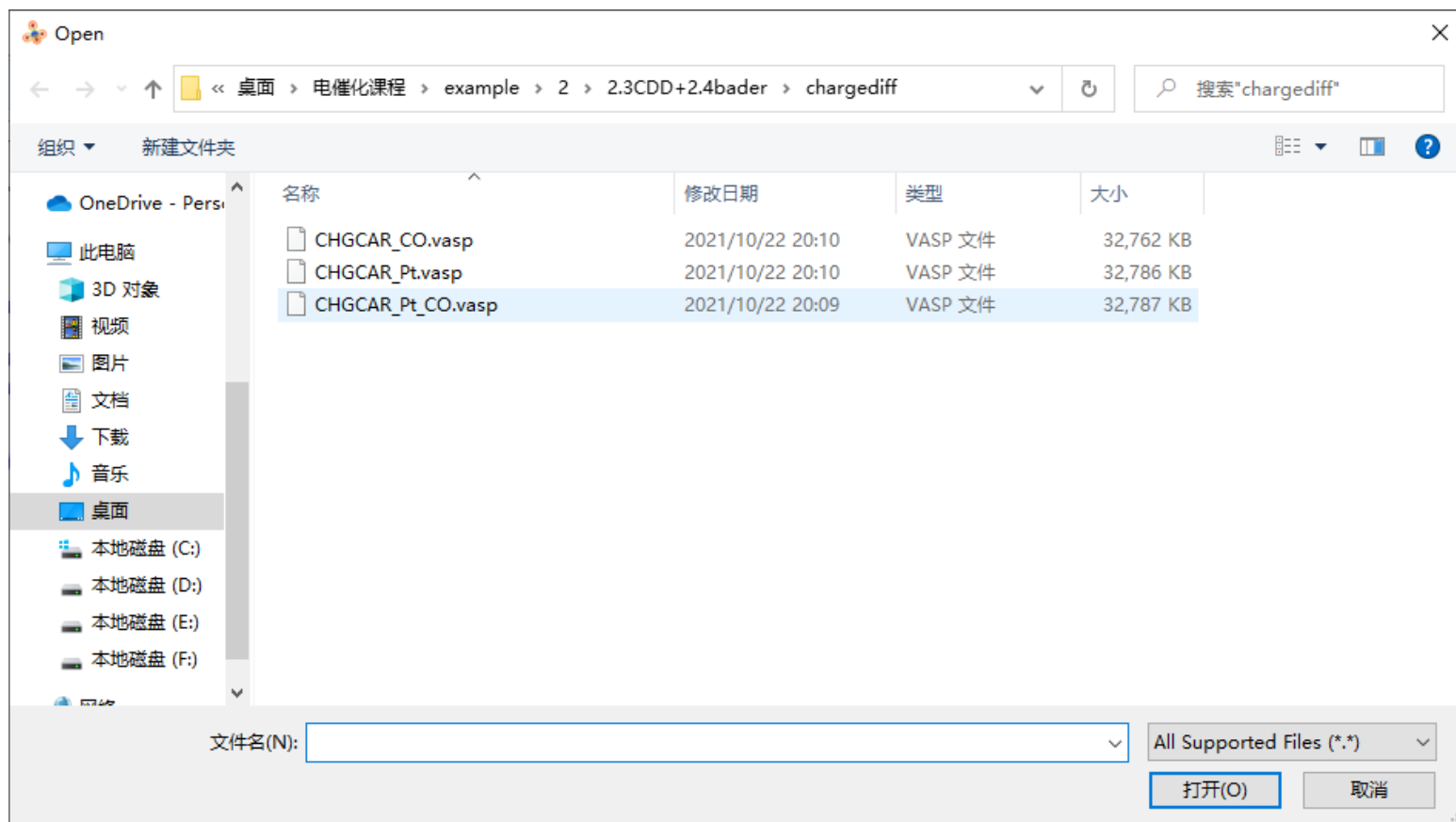
1. 下载计算后输出的CHGCAR_Pt_CO、CHGCAR_CO和CHGCAR_Pt 文件到桌面，将CHGCAR_Pt_CO 拖进 VESTA软件



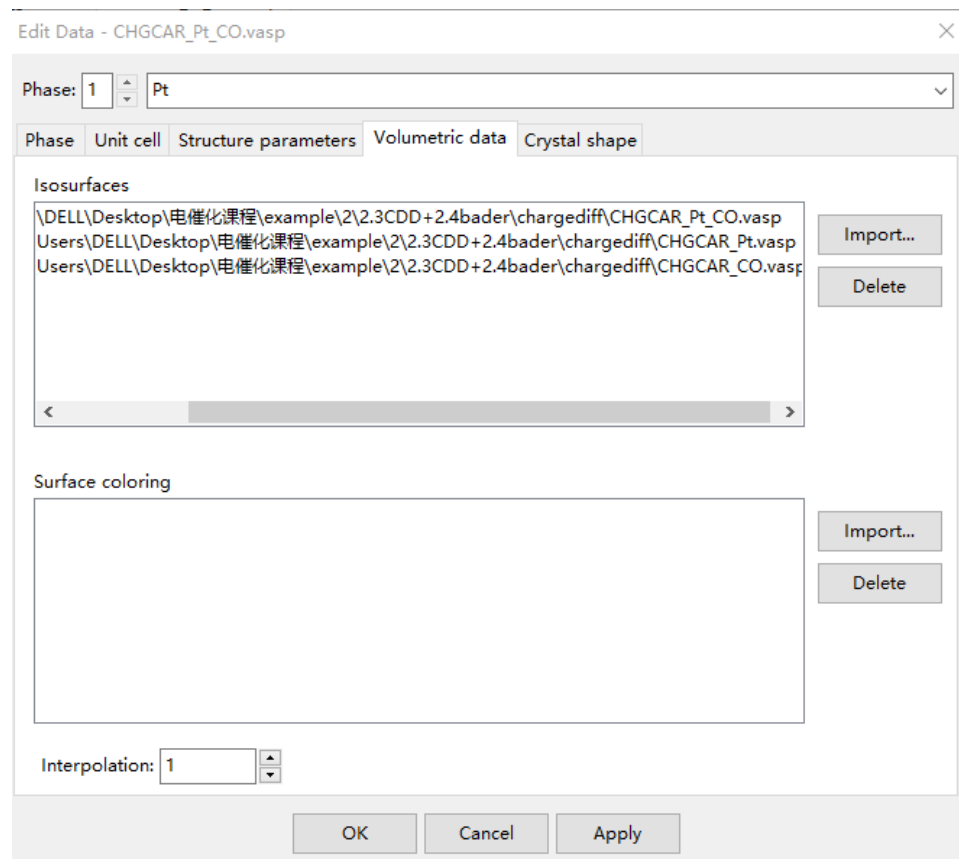
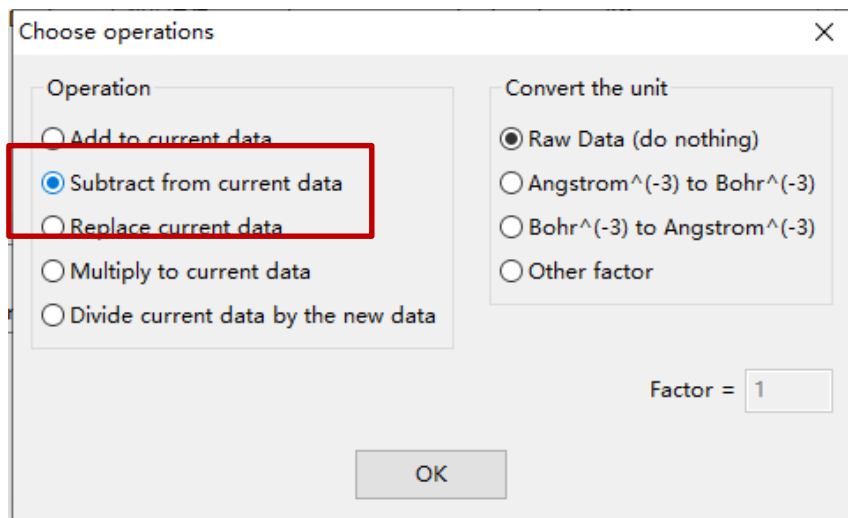
2. 点击Edit -- Edit Data -- Volumetric Data, 在新打开的窗口中点击Import



3. 依次选择文件夹里的CHGCAR_CO, CHGCAR_Pt 文件点击Open进行添加



4. 选择 **Select Subtract from current data**, 点击 **OK** 按钮, 即可将 **CHGCAR_CO** 和 **CHGCAR_Pt** 导入



5. 选择Properties-- isosurface修改等值面等进行美化

Properties - CHGCAR_Pt_CO.vasp

General Atoms Bonds Polyhedra **Isosurfaces** Sections

Material
Specular: 0 0 0 Shininess (%): 100

Isosurfaces
F(min) = -0.0828542; F(max) = 0.0192452;
☐ Render from front to back
Positive and negative Opacity 1 (0~255): 0
Isosurface level: 1 Opacity 2 (0~255): 0
Color: 0 255 255

No.	level	mode	color
1	0...0968	Positive and negative	
2	0.001	Negative	

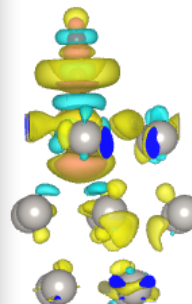
New
Delete
Clear

Surface coloring
F(min) = 0.000000; F(max) = 0.000000;
☒ Inverse color
Saturation level
Max.: 0.00370968 (100 %)
Min.: -0.00370968 (0 %)

☒ Preview

OK Cancel Save as Default

Step (px): 10 Step (%): 10



786 (715)
634 (475)
464 (477)

=====
isosurface = 108012 (54614)
= 71 ms

Properties...
Output Comment

电催化计算实战入门

VASP实操4—bader计算

主讲人：高怡静

差分电荷是对电荷转移情况定性的分析，bader是定量的分析。两者均是基于结构优化完成的基础上。

```
SYSTEM =Pt

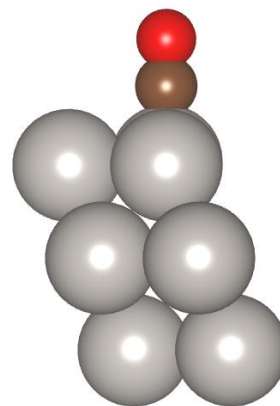
#Startparameter for this run:
PREC    = A
ISTART  = 0
ICHARG  = 2

ENCUT   = 500
NELM    = 1000; #maximum of electronic step
EDIFF   = 1E-5
LREAL   = Auto
EDIFFG  = -1E-2
NSW     = 0  # number of ioinc step
IBRION  = -1  # optimization
ISIF    = 2
ISMEAR  =0
SIGMA   = 0.2
LCHARG=T
LWAVE=T
GGA=PE
IJDW=11
LAECHG=T
```

关键词**LAECHG**

Automatic mesh

```
0
G
3   3   1
0.  0.  0.
```



Bader电荷处理，获得ACF文件。

计算结果

 AECCAR0

 AECCAR1

 AECCAR2

 CHG

 CHG CAR

 CONTCAR

 DOSCAR

 EIGENVAL

 IBZKPT

 INCAR

 KPOINTS

 log

 OSZICAR

 OUTCAR

处理过程

1. chgsum.pl AECCAR0 AECCAR1 生成CHARGE_sum

```
[cailiaoren1@iZuf6ewt8qjkzelxd4lyr9Z 1]$ chgsum.pl
usage: chgsum.pl <CHGCAR1> <CHGCAR2> <fact1> <fact2>
[cailiaoren1@iZuf6ewt8qjkzelxd4lyr9Z 1]$ chgsum.pl AECCAR0 AECCAR1
Atoms in file1: 2, Atoms in file2: 2
Points in file1: 1843200, Points in file2: 1843200
```

2. bader CHG CAR -ref CHG CAR_sum,得到ACF.dat

```
[cailiaoren1@iZuf6ewt8qjkzelxd4lyr9Z 1]$ bader CHG CAR -ref CHG CAR_sum

GRID BASED BADER ANALYSIS (Version 1.04 02/08/20)

OPEN ... CHG CAR
VASP5-STYLE INPUT FILE
DENSITY-GRID: 80 x 80 x 288
CLOSE ... CHG CAR
RUN TIME: 0.39 SECONDS
```

ACF文件

#	X	Y	Z	CHARGE	MIN DIST	ATOMIC VOL
1	1.387300	0.000000		2.718506	9.924389	1.265846
2	1.393978	1.597772		5.105924	9.856654	0.998150
3	-0.000028	0.800974		0.453117	10.016614	1.272057
4	4.161900	0.000000		2.718506	9.889151	1.265846
5	4.168717	1.600896		4.994500	10.062165	1.276832
6	2.774572	0.800974		0.453117	10.061422	1.272057
7	0.000000	2.402874		2.718506	9.983047	1.259227
8	0.004356	3.998450		4.994383	10.062605	1.275975
9	-1.387328	3.203848		0.453117	10.022143	1.272057
10	2.774600	2.402874		2.718506	9.978853	1.265846
11	2.782868	3.998949		4.994109	10.061699	1.285519
12	1.387272	3.203848		0.453117	10.075064	1.272057
13	1.403015	1.579601		6.947824	2.987128	0.330145
14	1.410990	1.567049		8.105225	7.019066	0.713735

这一列表示原子电荷，原子顺序和**POSCAR**排布顺序一样

分析处理

grep ZVAL POTCAR查看POTCAR中的Pt ,C,O原子中的原始电

```
[root@iZuf6ewt8qjkd41yr9Z 3]# grep ZVAL POTCAR
POMASS = 195.080; ZVAL = 10.000 mass and valenz Pt
POMASS = 12.011; ZVAL = 4.000 mass and valenz C
POMASS = 16.000; ZVAL = 6.000 mass and valenz O
```

与POTCAR中相应的原子价态相比较C的价电子数是4个，现在变成了6.94，说明这个C得到了 $6.94-4=2.94$ 个电子

与POTCAR中相应的原子价态相比较O的价电子数是6个，现在变成了8.11，说明这个O得到了 $8.11-6=2.11$ 个电子

因此这个CO从金属Pt得到了 $2.94+2.11=5.05$ 个电子

电催化计算实战入门

VASP实操5——态密度计算

主讲人：高怡静

态密度是每个轨道的电子云分布比例，通过态密度可以了解电子结构。同样需要结构优化完成

1. 静态计算

```
SYSTEM =Pt
```

```
#Startparameter for this run:
```

```
PREC   = A
```

```
ISTART = 0
```

```
ICHARG = 2
```

```
ENCUT  = 500
```

```
NELM   = 1000; #maximum of electronic step
```

```
EDIFF  = 1E-5
```

```
LREAL  = Auto
```

```
EDIFFG = -1E-2
```

```
NSW    = 0 # number of ioinc step
```

```
IBRION = -1 # optimization
```

```
ISIF   = 2
```

```
ISMEAR =0
```

```
SIGMA  = 0.2
```

```
LCHARG=T
```

```
LWAVE=T
```

```
GGA=PE
```

```
IVDW=11
```

```
LORBIT=11
```

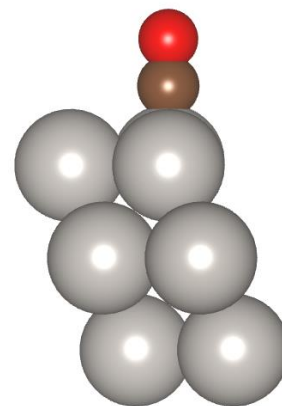
Automatic mesh

0

G

3 3 1

0. 0. 0.



2. 非自洽计算。将静态计算的CHG、CHGCAR和WAVECAR文件复制到新的文件夹，设置较大的K点,在INCAR中设置参数读取CHGCAR和WAVECAR文件

```
PREC   = A
ISTART = 1
ICHARG = 11
ISPIN=2
NEDOS=2000

ENCUT  = 500
NELM   = 1000; #maximum of electronic step
EDIFF  = 1E-5
LREAL  = Auto
EDIFFG = -1E-2
#NSW   = 1000  # number of ioinc step
#IBRION = 2    # optimization
ISIF   = 2
ISMEAR = -5
SIGMA  = 0.2
LORBIT=11
LCHARG=.TRUE.
LWAVE=.TRUE.
IVDW=11
GGA=PE
```

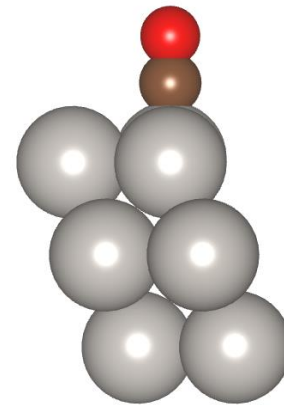
Automatic mesh

0

G

6 6 1

0. 0. 0.



vaspkit: 辅助处理态密度

1. 输入vaspkit

```
[cailiaoren1@iZuf6ewt8qjkelxd4lyr9Z dos2]$ vaspkit

      /\ /\ /\
     /  \  \
    (| (o) (o) |)      Hey, you must know what you are doing.
                        Otherwise you might get wrong results.
+-----oOoo--()--oOoo-----+
|                               VASPKIT Version 1.2.5 (08 Feb. 2021)|
|               Core Developer: Vei WANG (wangvei@icloud.com)|
|   Main Contributors: Nan XU, Jin-Cheng LIU & Gang TANG   |
|   Online Tutorials Available on Website: https://vaspkit.com |
+-----oooO-----+
|   ( )   Oooo.                               VASPKIT Made Simple
|   \ (   ( )
|   \_)  ) /
```

2. 输入11根据需要进行选择，此处选择115

```
11
===== DOS Options =====
111) Total Density-of-States
112) Projected Density-of-States of Only-One-Selected Atom
113) Projected Density-of-States of Each Element
114) Projected Density-of-States of Selected Atoms
115) Projected Density-of-States of Selected Atoms and Orbitals
116) Local Density-of-States of Each Element
117) Total Density-of-States from EIGENVAL File
118) Projected Density-of-States from EIGENVAL and PROCAR Files
119) Projected Density-of-States of Specified K-index
120) Projected Density-of-States of Specified Band-index

0)  Quit
9)  Back
```

3. 选择需要研究的原子

```
115
-->> (01) Reading Input Parameters From INCAR File...
+-----+
|          >>> The Fermi Energy will be set to zero eV <<<          |
+-----+
-->> (02) Reading Fermi-Energy from DOSCAR File...
-->> (03) Reading DOS Data From DOSCAR File...
-->> (04) Reading Structural Parameters from POSCAR File...
|-----|
The selected atoms will be listed in SELECTED_ATOMS_LIST file.
Input element-symbol and/or atom-index to choose [1 <= & <= 14]
(Free Format is Compatible, e.g., H O Au 1-4 7 18 all)

----->>
Pt
```

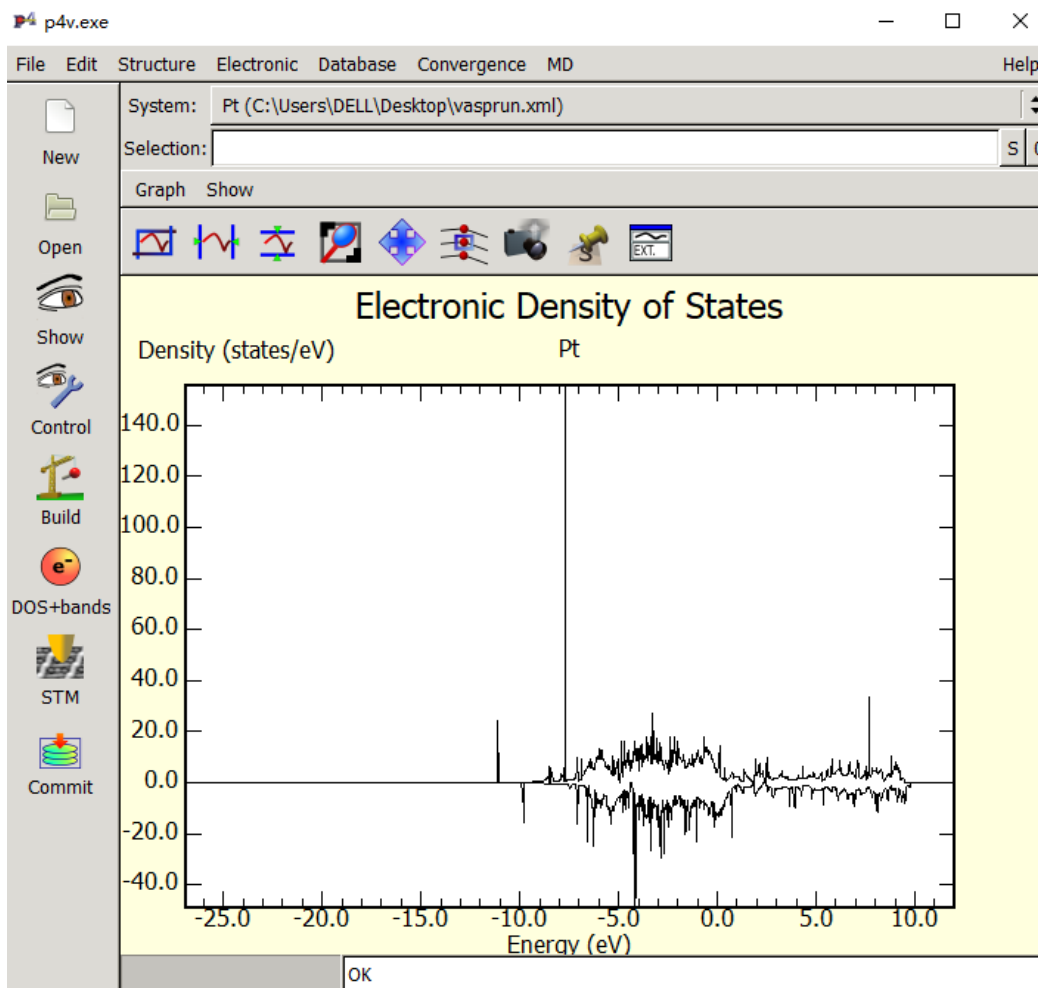
4. 选择需要研究的轨道

```
Selected ORBITALS will be listed in SELECTED_ORBITALS_LIST file.
Input the ORBITALS index to choose [1 <= & <= 9]
(Free format is compatible, e.g., 1-4 5 all)
1) s
2) py
3) pz
4) px
5) dxy
6) dyz
7) dz2
8) dxz
9) x2-y2

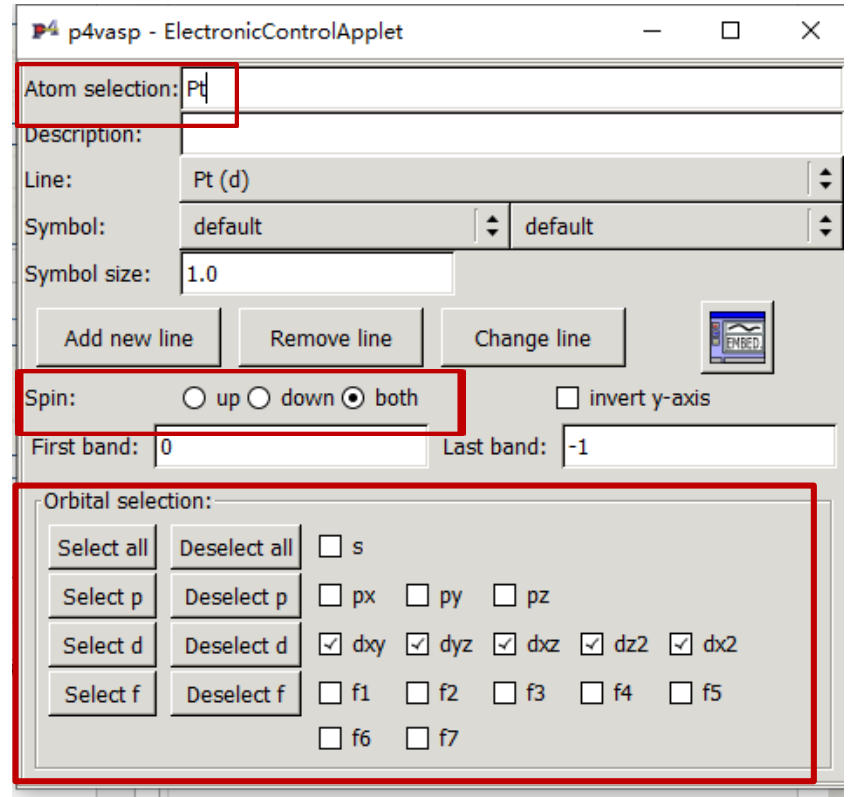
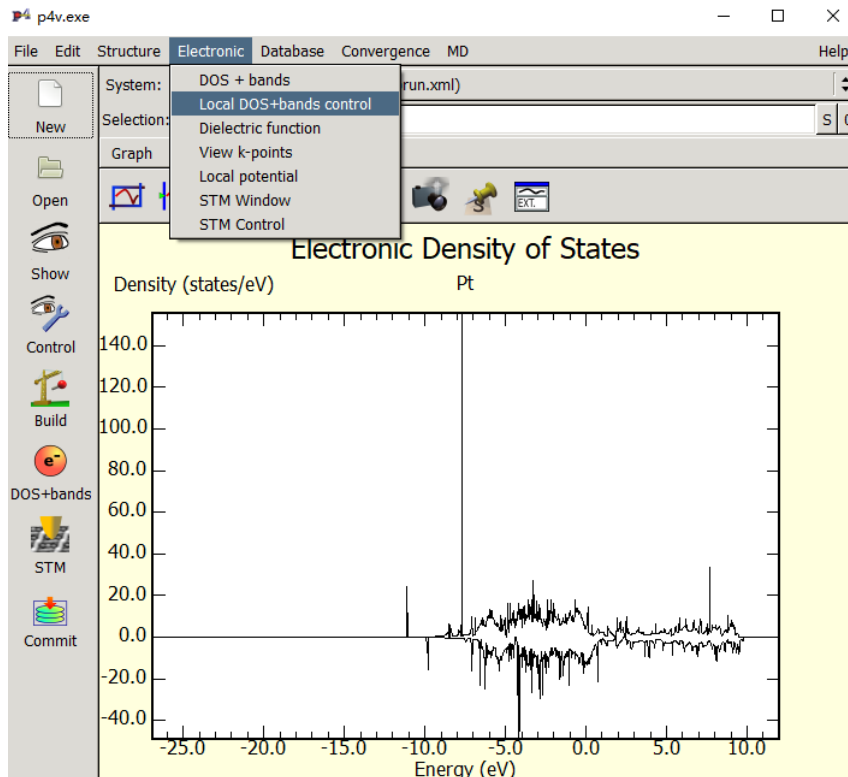
----->>
5
-->> (06) Written SELECTED_ORBITAS_LIST File!
-->> (07) Written PDOS_SUM.dat and IPDOS_SUM.dat Files!
```

态密度可视化

1. 将经过dos2的计算后输出**vasprun.xml**文件复制到桌面，并在桌面运行 **P4vasp.exe**软件，导入**vasprun.xml**文件。



2. 点击**Electronic – Local DOS+bands control**，输入所需研究的原子和所需研究的轨道。

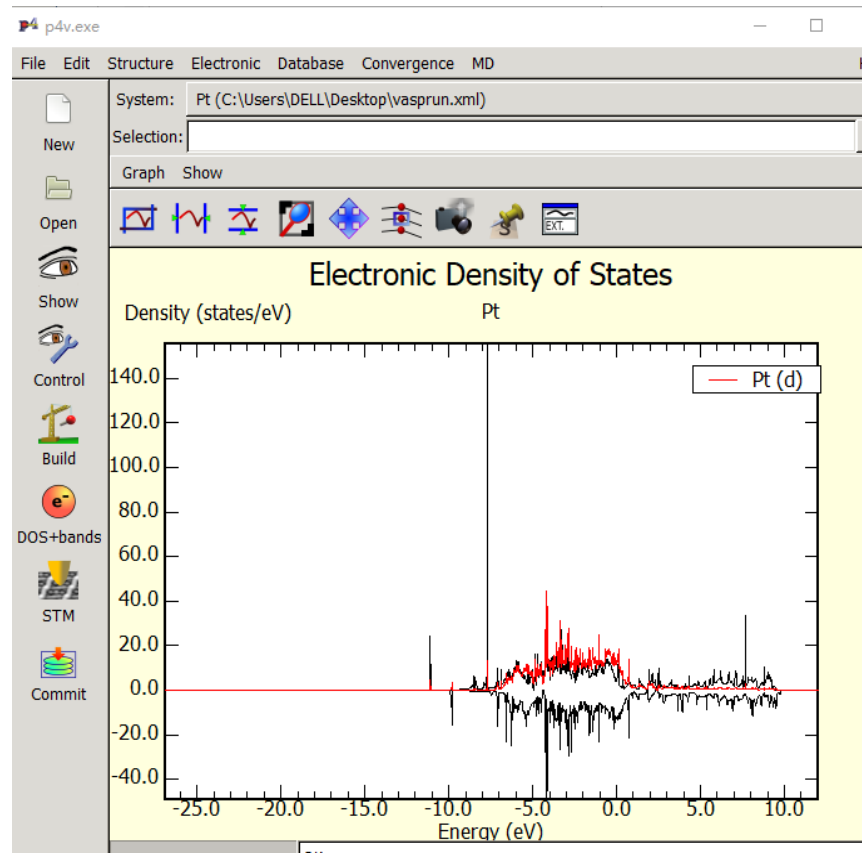
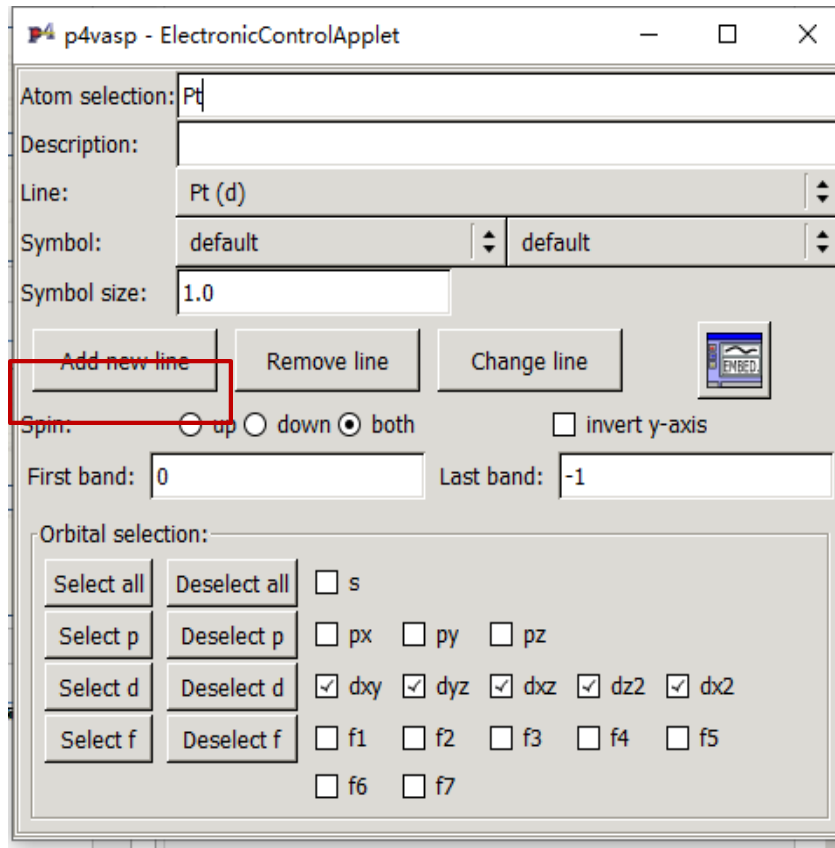


The figure shows the 'p4vasp - ElectronicControlApplet' window. The 'Atom selection' field is set to 'Pt'. The 'Description' field is empty. The 'Line' field is set to 'Pt (d)'. The 'Symbol' field is set to 'default'. The 'Symbol size' field is set to '1.0'. The 'Spin' section shows 'both' selected. The 'First band' field is set to '0' and the 'Last band' field is set to '-1'. The 'Orbital selection' section shows 'dxy', 'dyz', 'dxz', 'dz2', and 'dx2' selected.

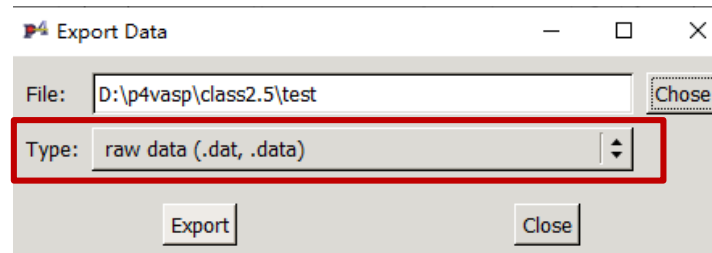
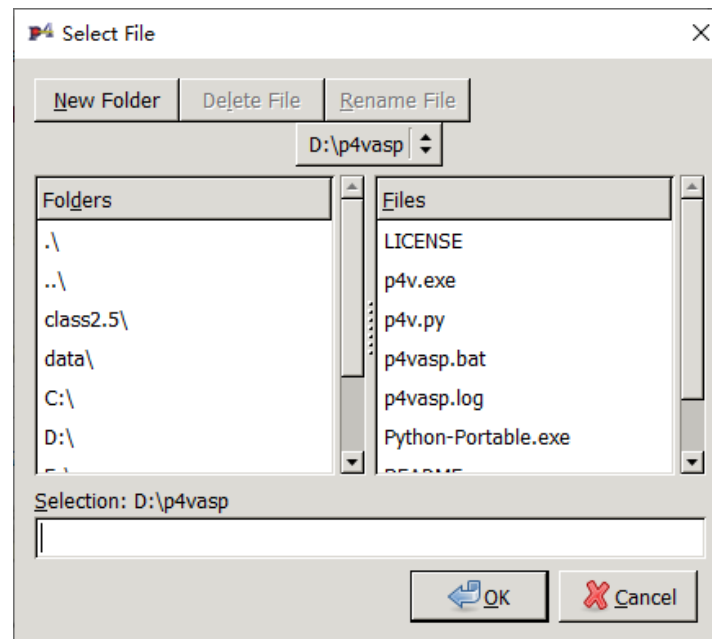
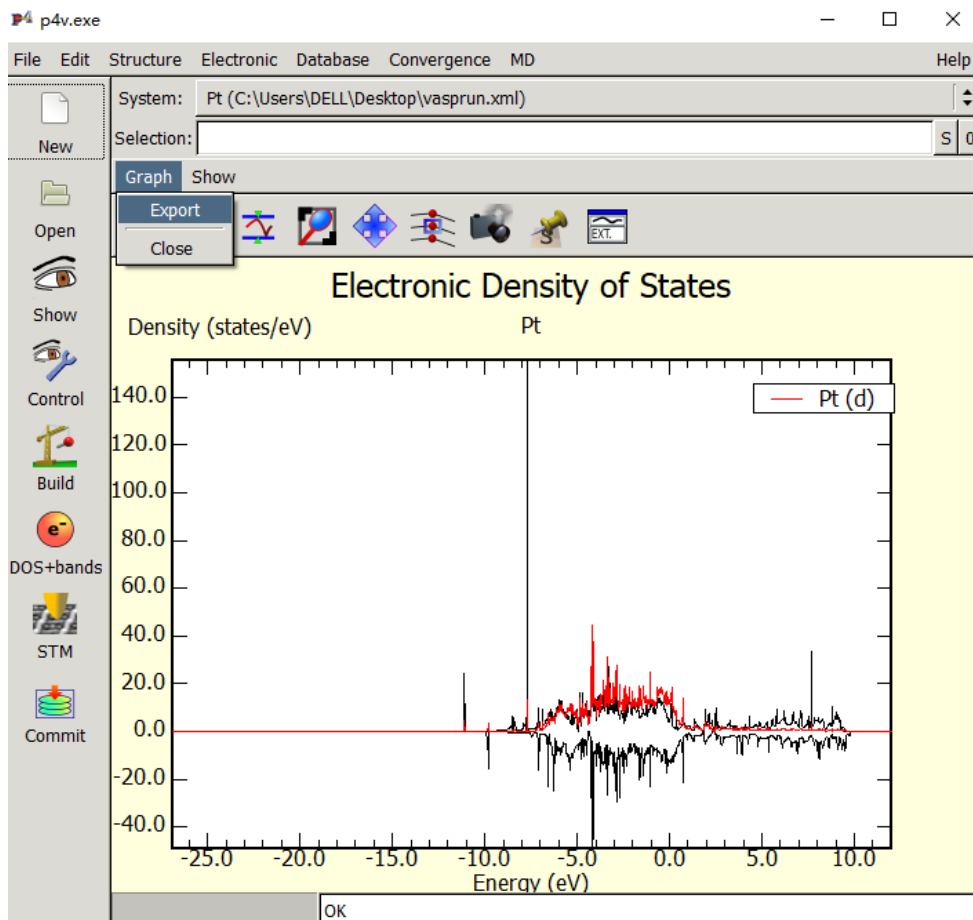
Orbital selection:	
Select all	Deselect all
Select p	Deselect p
Select d	Deselect d
Select f	Deselect f
☐ s	☐ px ☐ py ☐ pz
☒ dxy ☒ dyz ☒ dxz ☒ dz2 ☒ dx2	☐ f1 ☐ f2 ☐ f3 ☐ f4 ☐ f5
	☐ f6 ☐ f7

态密度可视化

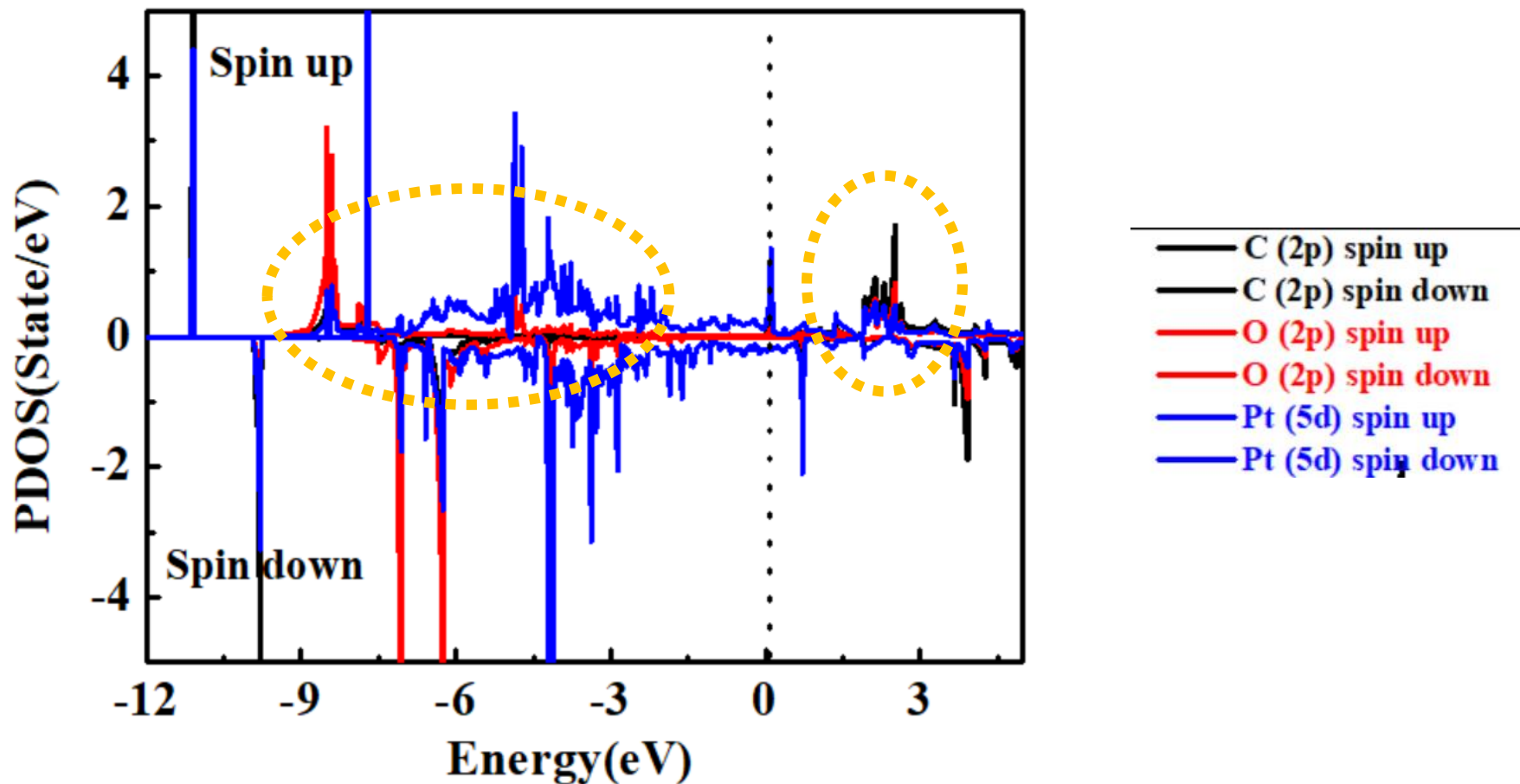
3. 点击add new line可以进行预览。



4. Graph--Export进行输出，指定输出路径



5. 结果分析



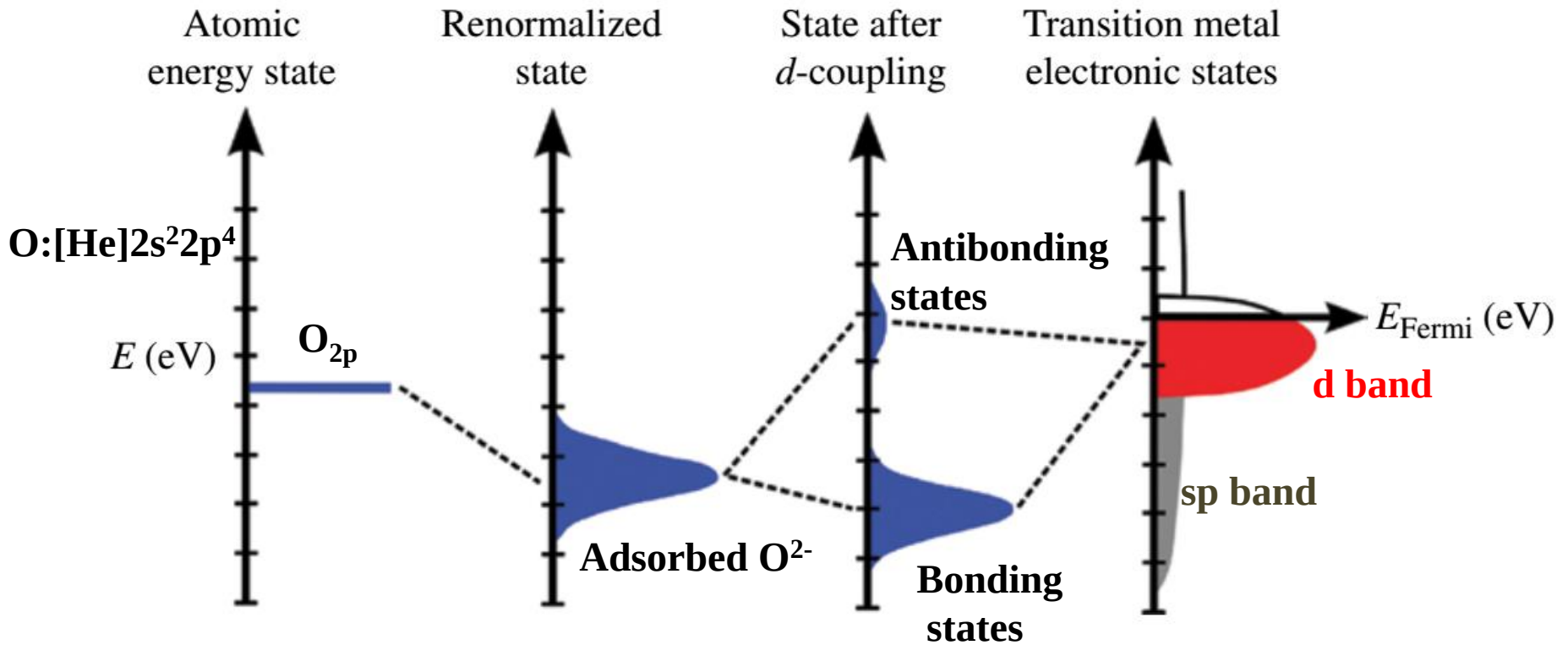
虚线代表的是费米等级，黄色区域中可以看出C_2p, O_2p和Pt_5d轨道重叠较多，表面CO和Pt之间有很强的电子转移。

电催化计算实战入门

VASP实操6— d 带中心

主讲人：高怡静

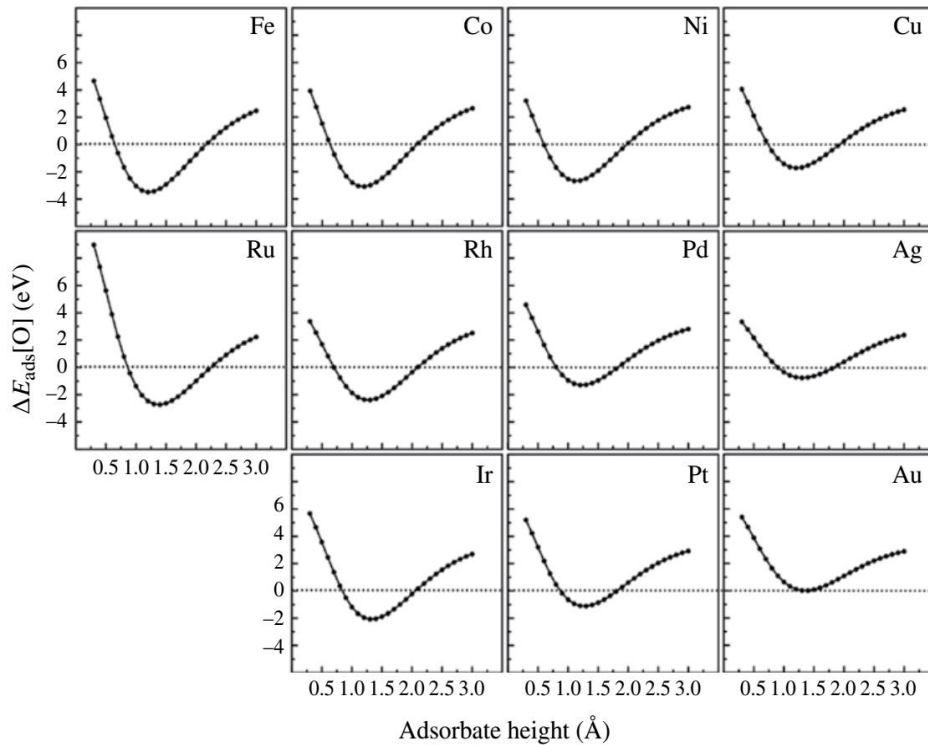
O原子在金属表面吸附



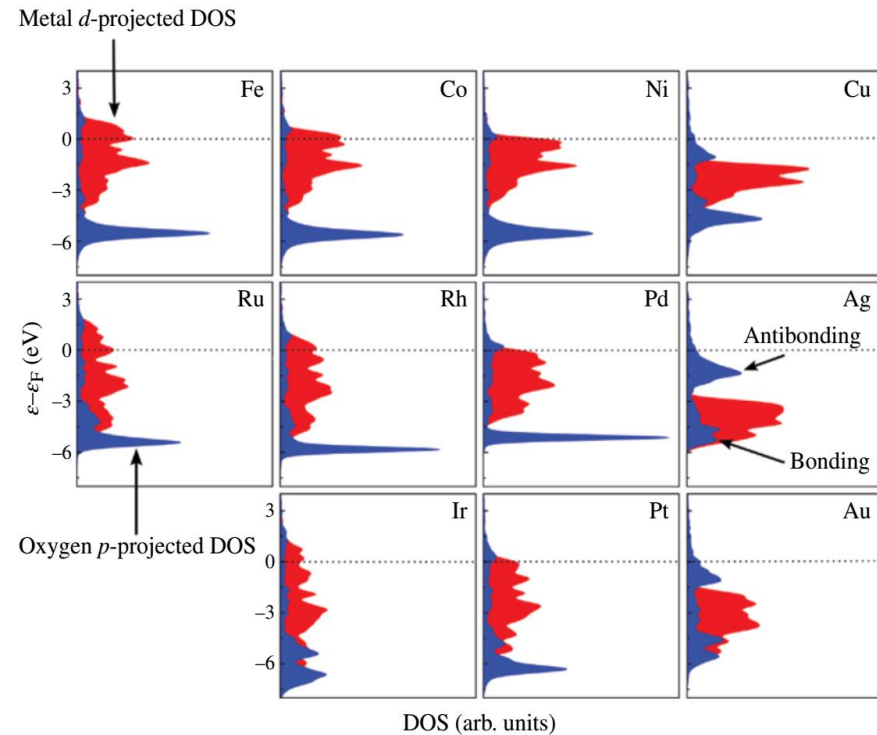
d 带中心越靠上，反键态填充位置越高，吸附能越大。

Jens K. Nørskov et.al Concepts in Heterogeneous Catalysis-Wiley (2014)

吸附能与距离的关系图



态密度图



从左到右，吸附能在逐渐减弱;从左到右， d 能级逐渐降低。

计算公式

The DOS projected onto the *d*-states that interact with the adsorbate state can be characterized by the moments of the *d* DOS. The first moment is the *d*-band center:

$$\varepsilon_d = \frac{\int_{-\infty}^{\infty} n_d(\varepsilon) \varepsilon d\varepsilon}{\int_{-\infty}^{\infty} n_d(\varepsilon) d\varepsilon}$$

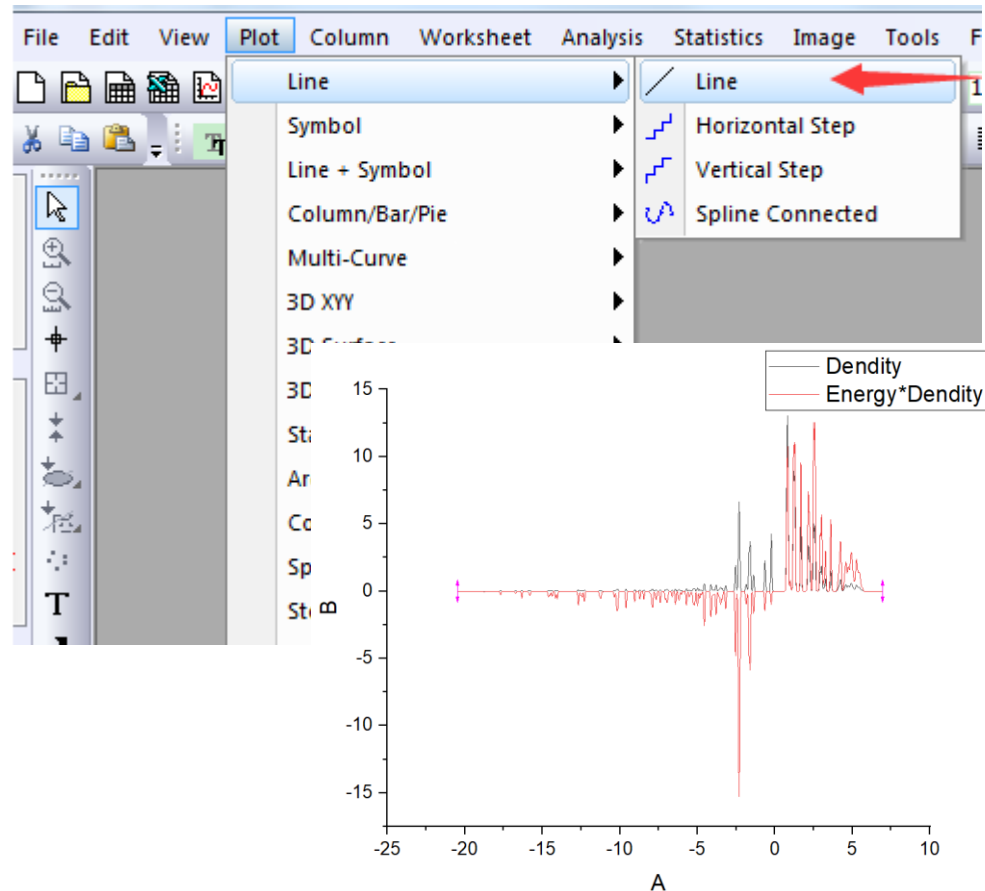
ε 为能量， $n_d(\varepsilon)$ 为电子的密度

举例：以Ti原子为例

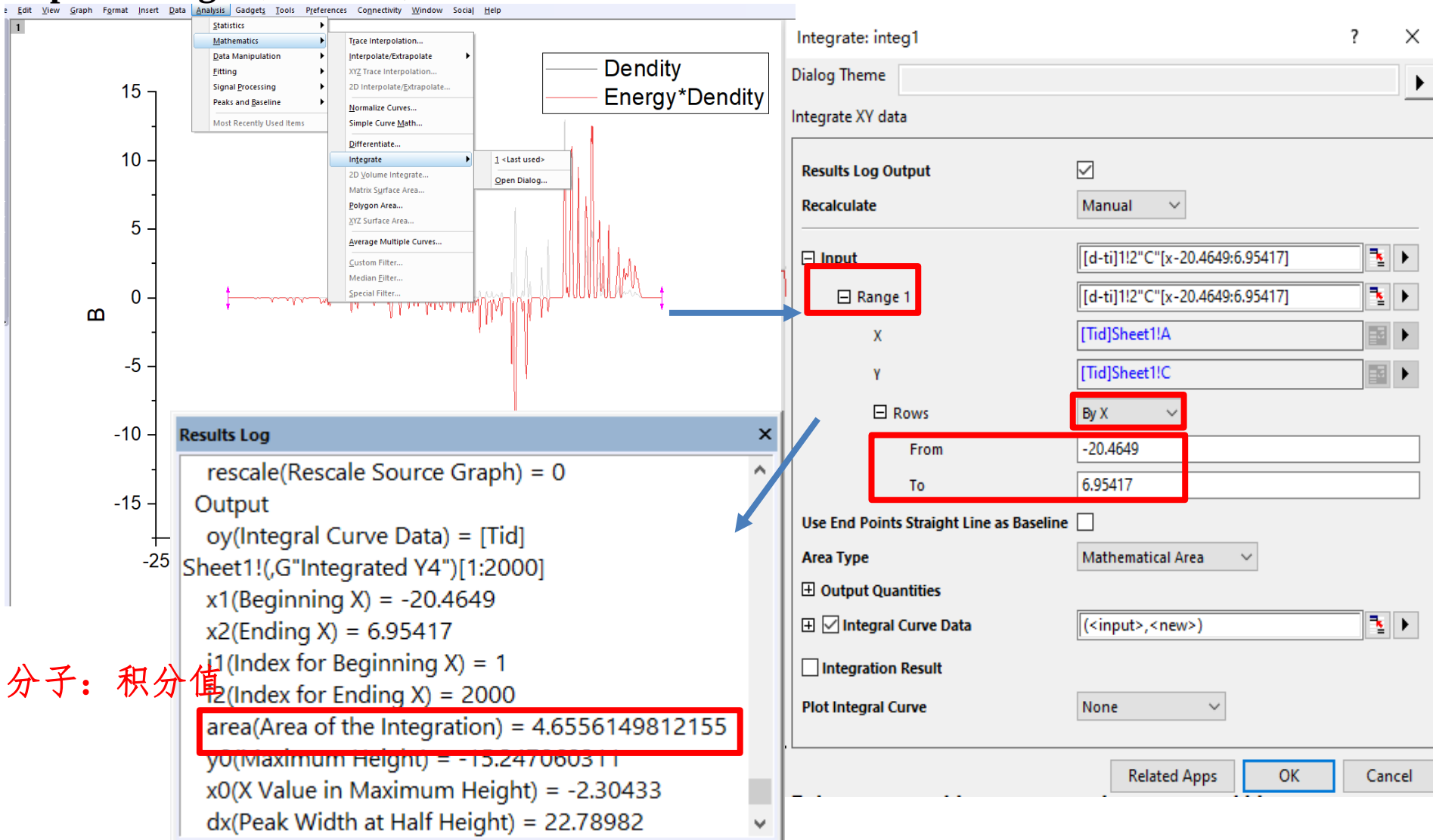
1. 将DOS可视化处理的文件中的能量态密度以及能量*态密度导入Origin

2. 点击 **plot --Line -- Line** 绘制线性图

	A(X)	B(Y)	C(Y)
Long Name			
Units			
Comments			
F(x)=	能量	态密度	能量*态密度
1	-20.4649	0	0
2	-20.4511	0	0
3	-20.4374	0	0
4	-20.4237	0	0
5	-20.41	0	0
6	-20.3963	0	0
7	-20.3826	0	0
8	-20.3688	0	0
9	-20.3551	0	0
10	-20.3414	0	0
11	-20.3277	0	0
12	-20.314	0	0
13	-20.3003	0	0
14	-20.2865	0	0
15	-20.2728	0	0
16	-20.2591	0	0
17	-20.2454	0	0
18	-20.2317	0	0
19	-20.218	0	0
20	-20.2042	0	0
21	-20.1905	0	0
22	-20.1768	0	0
23	-20.1631	0	0
24	-20.1494	0	0
25	-20.1357	0	0
26	-20.1219	0	0
27	-20.1082	0	0
28	-20.0945	0	0
29	-20.0808	0	0

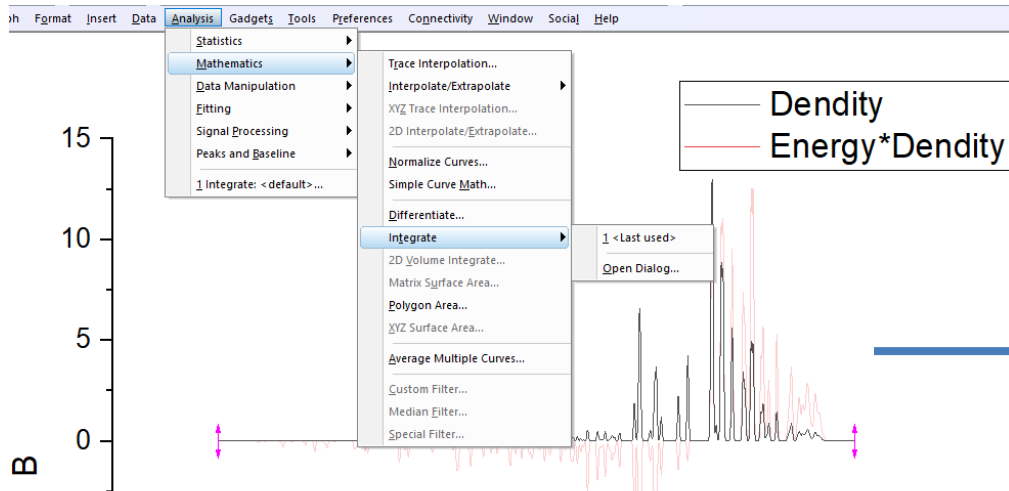


3. 能量*态密度求积分。选中红色曲线点击Analysis --Mathematics --Integrate --Open Dialog, 设置之后点击OK.



分子：积分值

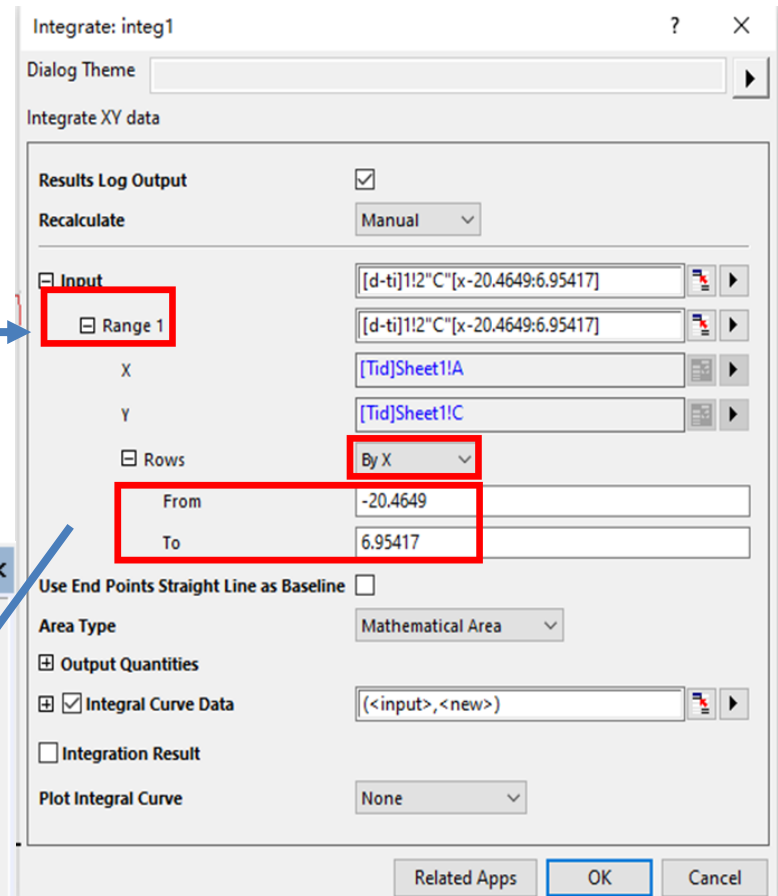
3. 态密度求积分。选中黑色曲线点击Analysis -> Mathematics -> Integrate -> Open Dialog, 设置之后点击OK



Results Log

Sheet1!(H"Integrated Y5")[1:2000]
x1(Beginning X) = -20.4649
x2(Ending X) = 6.95417
i1(Index for Beginning X) = 1
i2(Index for Ending X) = 2000
area(Area of the Integration) = 8.7676593983
y0(Maximum Height) = 13.0415
x0(X Value in Maximum Height) = 0.82297
dx(Peak Width at Half Height) = 3.6260949895054

分母：积分值



The dialog box 'Integrate: integ1' shows the 'Integrate XY data' section. The 'Input' section has 'Range 1' selected. The 'X' and 'Y' fields are both set to '[d-ti]!1/2"C"[x-20.4649:6.95417]'. The 'Rows' section is set to 'By X'. The 'From' field is '-20.4649' and the 'To' field is '6.95417'. The 'Area Type' is 'Mathematical Area'. The 'Output Quantities' section has 'Integral Curve Data' checked. The 'Plot Integral Curve' is set to 'None'. The 'OK' button is highlighted.

最后将两次积分结果相除，计算结果为： $4.656/8.768 = 0.531$

Thank you !